

Steady state error

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Mechatronics 3DX4

Slides 7: Steady-State Errors

Dr. O. Onaizah

Department of Computing and Software
McMaster University

Material based on lecture notes by P. Taylor and M. Lawford, and *Control Systems Engineering* by N. Nise.

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Introduction

- ▶ We now focus on the third design specification, *steady-state error*.

$$e_{ss} = \lim_{t \rightarrow \infty} e(t), \quad e(t) = r(t) - y(t)$$

- ▶ We define **steady-state error** to be the difference between input and output as $t \rightarrow \infty$.

- We will see that control system design typically means we will have to make trade-offs between the desired transient, steady-state, and stability specifications.

Test Inputs

- Table below shows the standard test inputs typically used for evaluating steady-state error.

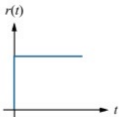
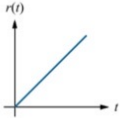
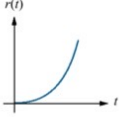
Waveform	Name	Physical interpretation	Time function	Laplace transform
	Step	Constant position	1	$\frac{1}{s}$
	Ramp	Constant velocity	t	$\frac{1}{s^2}$
	Parabola	Constant acceleration	$\frac{1}{2}t^2$	$\frac{1}{s^3}$

Table 7.1.

Choosing a Test Inputs

- The test inputs we will choose for our steady-state analysis and design depends on our target application.

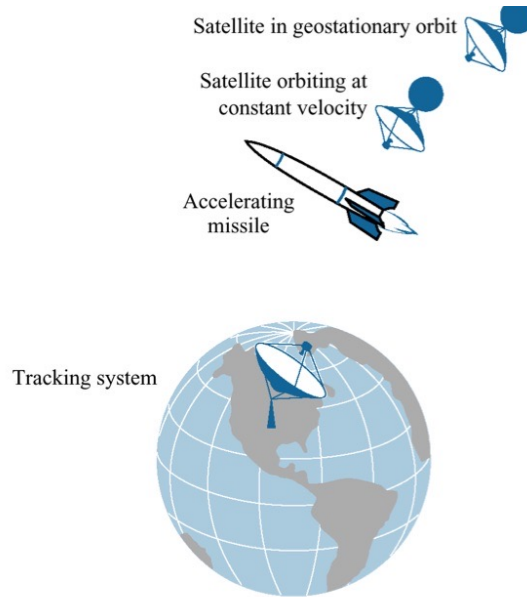


Figure 7.1.

Steady-State Error and Stable Systems

- ▶ The calculations we will be deriving for steady-state apply only to stable systems.
- ▶ Unstable systems represent loss of control in steady-state as the transient response swamps the forced response.
- ▶ As we analyze and design a system for steady-state error, we must constantly check the system for stability.

Steady-State Error and Step Inputs

- ▶ With step inputs, we can get two types of steady-state errors:
 1. Zero error.
 2. A constant error value.

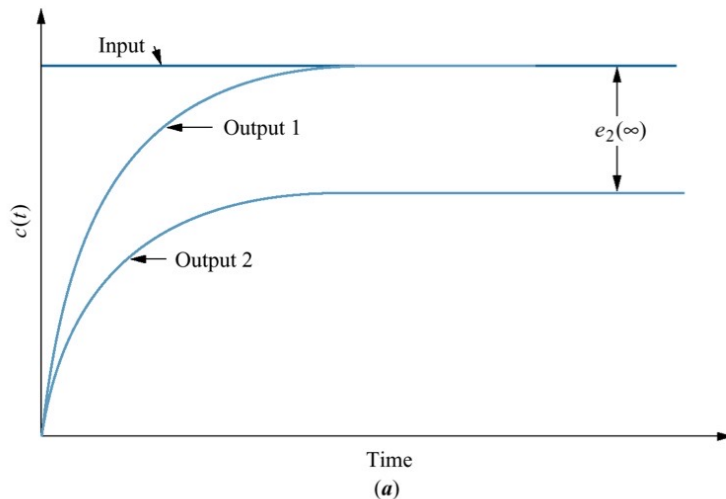
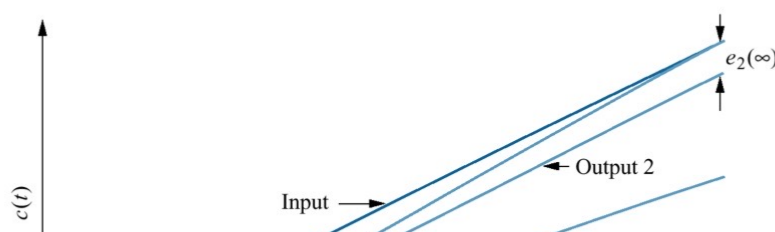


Figure 7.2.

对于step response, 系统一般有两种响应: zero error, 即响应和输入完全没有差距; 或者constant error, 响应和输入仍差一个固定值。

Steady-State Error and Ramp input

- ▶ With ramp inputs, we can get three types of steady-state errors:
 1. Zero error.
 2. A constant error value.
 3. Infinite error.



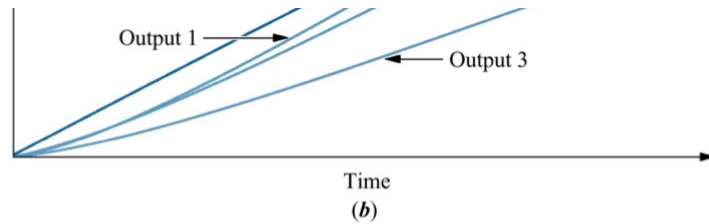


Figure 7.2.

对于ramp输入，有三种可能：zero error, a constant error value, infinite error.

Steady-State Error and Block Diagrams

- ▶ If we have a closed-loop transfer function $T(s)$, we can represent our error signal, $E(s)$, as in figure (a).
- ▶ We are interested in the time domain signal, $e(t) = \mathcal{L}^{-1}\{E(s)\}$, as $t \rightarrow \infty$.
- ▶ If we have a unity feedback system, we already have $E(s)$ as part of our diagram, as shown in figure (b).

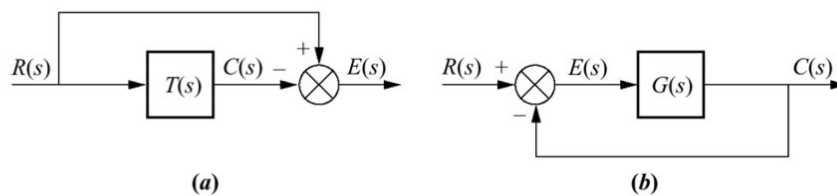


Figure 7.3.

注：T是闭环传递函数，即 $T(s) = \frac{G(s)}{1+G(s)*H(s)}$

如何理解图a：E(s)就是error在频域的表达形式，即输入R(s)-响应C(s),
 $E(s) = R(s) - C(s) = R(s)(1 - T(s))$

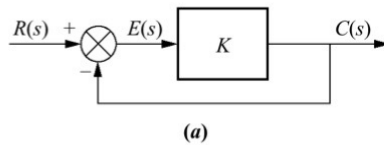
图b就是用标准的名后块状太图告诉我们系统的关系 $E(s) = R(s) / (1 + G(s)H(s))$

Sources of Steady-State Error

- ▶ Steady-state errors can arise from nonlinear sources, such as backlash in gears or motors requiring a minimum input voltage before it starts to move.
- ▶ Steady-state errors can also arise from configuration of system and the input we apply.
- ▶ Consider a step input applied to the system below which has constant gain.
- ▶ If a unity feedback system has a feedforward transfer function $G(s)$, then we can derive the transfer function $\frac{E(s)}{R(s)}$ as follows:

$$C(s) = E(s)G(s) \quad (1)$$

$$E(s) = R(s) - C(s) \quad (2)$$



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Figure 7.4.

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误差的来源有几条：

1. 非线性因素，比如backlash，或者电压需要一个最小值驱动。
2. 有的系统会根据输入类型不同，steady state error不同。
3. 例子

Sources of Steady-State Error - II

- ▶ Substituting equation 1 into equation 2 gives:

$$\begin{aligned} E(s) &= R(s) - E(s)G(s) \\ E(s)[1 + G(s)] &= R(s) \\ \frac{E(s)}{R(s)} &= \frac{1}{1 + G(s)} \end{aligned} \quad (3)$$

- For $G(s) = K$, we get

$$\frac{E(s)}{R(s)} = \frac{1}{1 + K} \quad (4)$$

- For $R(s) = \frac{1}{s}$ (unit step), we get $E(s) = \frac{1}{s(1+K)}$.

- We thus have $e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = \frac{1}{1+K}$

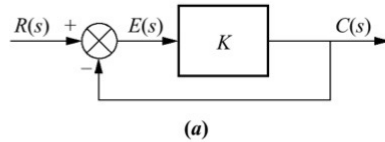


Figure 7.4.

Sources of Steady-State Error - III

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)}$$

- If we add an integrator to the forward-path gain, we get $G(s) = \frac{K}{s}$ giving

$$\frac{E(s)}{R(s)} = \frac{1}{1 + \frac{K}{s}} = \frac{s}{s + K} \quad (5)$$

- For $R(s) = \frac{1}{s}$ (unit step), we get $E(s) = \frac{1}{(s+K)}$.

- We thus have

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = \frac{0}{0 + K} = 0 \quad (6)$$

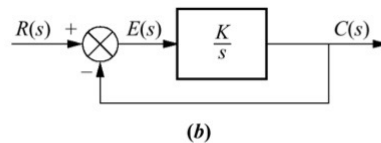


Figure 7.4.

Steady-State Error and $T(s)$

- ▶ In Diagram below, we have $E(s) = R(s) - C(s)$.
- ▶ We also have:

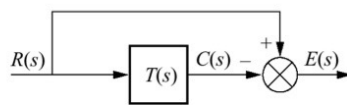
$$C(s) = R(s)T(s) \quad (7)$$

- ▶ Combining the two we get

$$E(s) = R(s) - R(s)T(s) = R(s)[1 - T(s)] \quad (8)$$

- ▶ We thus have

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s E(s) \\ &= \lim_{s \rightarrow 0} s R(s)[1 - T(s)] \end{aligned} \quad (9)$$



(a)

Figure 7.3.

Steady-State Error and $G(s)$

- ▶ From equation 3, we have

$$E(s) = \frac{R(s)}{1 + G(s)} \quad (10)$$

- ▶ We thus have

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s E(s) \\ &= \lim_{s \rightarrow 0} s \frac{R(s)}{1 + G(s)} \end{aligned} \quad (11)$$



(b)

Figure 7.3.

Steady-State Error, $G(s)$, and Step Input

- ▶ For input $R(s) = \frac{1}{s}$, we get

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{1/s}{1 + G(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)} \quad (12)$$

- ▶ We refer to the term $\lim_{s \rightarrow 0} G(s)$ as **dc gain of the forward transfer function**.
- ▶ To have zero steady-state error we need

$$\lim_{s \rightarrow 0} G(s) = \infty \quad (13)$$

- ▶ For $G(s)$ of form below, we thus need $n \geq 1$

$$G(s) \equiv \frac{(s + z_1)(s + z_2) \cdots}{s^n(s + p_1)(s + p_2) \cdots} \quad (14)$$

- ▶ If $n = 0$, we get

$$\lim_{s \rightarrow 0} G(s) = \frac{(0 + z_1)(0 + z_2) \cdots}{(0 + p_1)(0 + p_2) \cdots} = \frac{z_1 z_2 \cdots}{p_1 p_2 \cdots} \quad (15)$$

这里说的是，在之前的negative feedback system，当输入是step input的时候，G需要满足什么特性才让 $e_{ss}=0$ 。

Steady-State Error, $G(s)$, and Ramp Input

- ▶ For input $R(s) = \frac{1}{s^2}$, we get

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)} \quad (16)$$

- ▶ To have zero steady-state error for ramp input, we need

$$\lim_{s \rightarrow 0} sG(s) = \infty \quad (17)$$

- For $G(s)$ of form below, we thus need $n \geq 2$

$$G(s) \equiv \frac{(s + z_1)(s + z_2) \cdots}{s^n(s + p_1)(s + p_2) \cdots} \quad (18)$$

- If $n = 1$, we get

$$\lim_{s \rightarrow 0} s G(s) = \frac{z_1 z_2 \cdots}{p_1 p_2 \cdots} \quad (19)$$

- If $n = 0$, we get

$$\lim_{s \rightarrow 0} s G(s) = \frac{s(s + z_1)(s + z_2) \cdots}{(s + p_1)(s + p_2) \cdots} = 0 \quad (20)$$

这里说的是ramp input的情况。

Steady-State Error, $G(s)$, and Parabolic Input

- For input $R(s) = \frac{1}{s^3}$, we get

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s(1/s^3)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)} = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)} \quad (21)$$

- To have zero steady-state error for a parabolic input, we need

$$\lim_{s \rightarrow 0} s^2 G(s) = \infty \quad (22)$$

- For $G(s)$ of form below, we thus need $n \geq 3$

$$G(s) \equiv \frac{(s + z_1)(s + z_2) \cdots}{s^n(s + p_1)(s + p_2) \cdots} \quad (23)$$

- If $n = 2$, we get

$$\lim_{s \rightarrow 0} s^2 G(s) = \frac{z_1 z_2 \cdots}{p_1 p_2 \cdots} \quad (24)$$

- If $n = 1$, we get

$$\lim_{s \rightarrow 0} s^2 G(s) = \frac{s(s + z_1)(s + z_2) \cdots}{(s + p_1)(s + p_2) \cdots} = 0 \quad (25)$$

这里说的是parabolic input的情况。

Steady-State Error eg.

- Find the steady state errors for inputs $5u(t)$, $5tu(t)$, and $\frac{5}{2}t^2u(t)$.

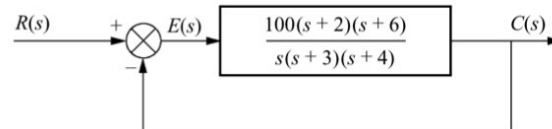


Figure 7.6.

$$r_1(t) = 5u(t) \Rightarrow R_1(s) = \frac{5}{s}$$

$$r_2(t) = 5tu(t) \Rightarrow R_2(s) = \frac{5}{s^2}$$

$$r_3(t) = \frac{5}{2}t^2u(t) \Rightarrow R_3(s) = \frac{5}{s^3}$$

For $R_1(s)$:

$$\begin{aligned} e_{ss1} &= 5 \left(\frac{1}{1 + \lim_{s \rightarrow 0} G(s)} \right) = 5 \left(\frac{1}{1 + \lim_{s \rightarrow 0} \frac{100(s+2)(s+6)}{s(s+3)(s+4)}} \right) \\ &= 5 \left(\frac{1}{1 + \frac{100 \times 2 \times 6}{0 \times 3 \times 4}} \right) \end{aligned}$$

For $R_2(s)$

$$e_{ss2} = 5 \left(\frac{1}{\lim_{s \rightarrow 0} s G(s)} \right) = 5 \left(\frac{1}{\lim_{s \rightarrow 0} s \frac{100(s+2)(s+6)}{s(s+3)(s+4)}} \right)$$

$$= 5 \left(\frac{1}{\frac{100 \times 2 \times 6}{3 \times 4}} \right) = 5 \times \frac{12}{1200} = 0.05$$

For $R_3(s)$

$$e_{ss3} = 5 \left(\lim_{s \rightarrow 0} \frac{1}{s^2 G(s)} \right) = 5 \left(\lim_{s \rightarrow 0} s^2 \frac{100(s+2)(s+6)}{s(s+3)(s+4)} \right)$$

$$= \infty$$

Static Error Constants

- We now define steady state-error performance specifications called **static error constants**.

1. Position Constant: $K_p = \lim_{s \rightarrow 0} G(s)$, thus

$$e_{step}(\infty) = \frac{1}{1 + K_p}$$

2. Velocity Constant: $K_v = \lim_{s \rightarrow 0} sG(s)$, thus

$$e_{ramp}(\infty) = \frac{1}{K}$$

3. Aceleration Constant: $K_a = \lim_{s \rightarrow 0} s^2 G(s)$, thus

$$e_{parabola}(\infty) = \frac{1}{K_a}$$

这里就是对标准负反馈系统的steady state error的简化公式，这样我们就不用每次都计算 $\lim_{s \rightarrow 0} s * E(s)$

System Type

- ▶ The static error constants are determined by the structure of $G(s)$.
- ▶ They are mostly determined by the number of integrators in $G(s)$.
- ▶ The **system type** is the number of integrators in the forward path, thus the value of n in figure below.

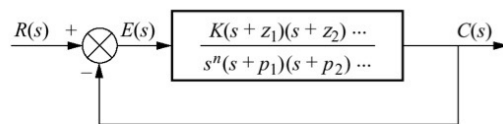


Figure 7.8.

System type 就是forward path的积分器个数，即G(s)分母s项的n次方。

Steady-State Error Summary

- Table shows relationship between input type, system type, static error constants, and steady-state errors.

Input	Steady-state error formula	Type 0		Type 1		Type 2	
		Static error constant	Error	Static error constant	Error	Static error constant	Error
Step, $u(t)$	$\frac{1}{1 + K_p}$	$K_p = \text{Constant}$	$\frac{1}{1 + K_p}$	$K_p = \infty$	0	$K_p = \infty$	0
Ramp, $tu(t)$	$\frac{1}{K_v}$	$K_v = 0$	∞	$K_v = \text{Constant}$	$\frac{1}{K_v}$	$K_v = \infty$	0
Parabola, $\frac{1}{2}t^2u(t)$	$\frac{1}{K_a}$	$K_a = 0$	∞	$K_a = 0$	∞	$K_a = \text{Constant}$	$\frac{1}{K_a}$

Table

7.2.

Tight Steady-State Error Specifications

- Example of a system requiring tight steady-state error specifications to be useful.



Figure 7.9.

Steady-State Error Specifications eg.

- For system below, find value of K such there is 10% error in steady state.

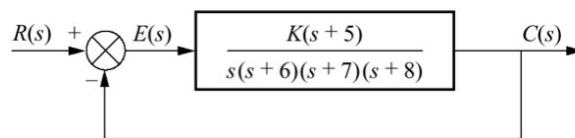


Figure 7.10.

Note: s^1 so type 1

For ramp input $R(s) = \frac{1}{s^2}$, then the error in steady state is

$$e_{ss} = \frac{1}{k_v}, \quad k_v = \lim_{s \rightarrow 0} s G(s) = \lim_{s \rightarrow 0} \frac{s K(s+5)}{s(s+6)(s+7)(s+8)}$$

$$= \frac{5K}{6 \times 7 \times 8} = \frac{5K}{336}$$

For ramp input, want e_{ss} to be 10%

$$e_{ss} = \frac{1}{K_v} = 0.1$$

$$\frac{1}{K_v} = \frac{1}{\frac{5K}{336}} = \frac{336}{5K} = 0.1$$

$$K = \frac{336}{5 \times 0.1} = 672$$

Must check that closed loop system is stable for $K = 672$

$$G_{CL}(s) = \frac{G(s)}{1 + G(s)}$$

All poles in LHP

Steady-State Error and Disturbances

- Can use feedback systems to handle unwanted disturbances to the systems.
- By using feedback, we can design systems that follow the input signal with small or zero error, *despite* these disturbances.

- Consider feedback system below with disturbance, $D(s)$, added between plant and controller.
- The system output is

$$C(s) = E(s)G_1(s)G_2(s) + D(s)G_2(s) \quad (26)$$

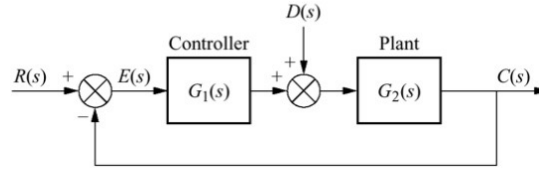


Figure 7.11.

有一些系统会有干扰输入，我们叫做disturbance，其也有可能造成error。这里G1是控制器，G2是被控制对象。

Steady-State Error and Disturbances - II

- However

$$E(s) = R(s) - C(s) \Rightarrow C(s) = R(s) - E(s) \quad (27)$$

- Using Equations 27 and 26 and solving for $E(s)$ gives

$$E(s) = \frac{R(s)}{1 + G_1(s)G_2(s)} - \frac{D(s)G_2(s)}{1 + G_1(s)G_2(s)} \quad (28)$$

- Using final-value theorem, the steady-state error is

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) \quad (29)$$

$$= \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G_1(s)G_2(s)} - \lim_{s \rightarrow 0} \frac{sD(s)G_2(s)}{1 + G_1(s)G_2(s)} \quad (30)$$

$$= e_R(\infty) + e_D(\infty) \quad (31)$$

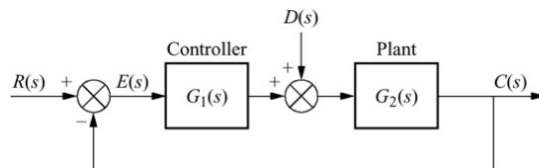


Figure 7.11.

Steady-State Error and Disturbances - III

- ▶ The $e_R(\infty)$ term is the steady-state error due to input $R(s)$ that we have already seen.
- ▶ The $e_D(\infty)$ term is the steady-state error due to $D(s)$.
- ▶ If $D(s) = 1/s$ (step input), we have

$$e_D(\infty) = -\frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)} \quad (32)$$

- ▶ If we set $R(s) = 0$, we get from Eqn28 the transfer function:

$$\frac{E(s)}{D(s)} = -\frac{G_2(s)}{1 + G_1(s)G_2(s)} \quad (33)$$

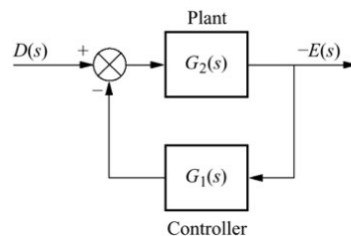


Figure 7.12.

我们这里让 $R(s)=0$ ，只研究disturbance $D(s)$ 的影响。

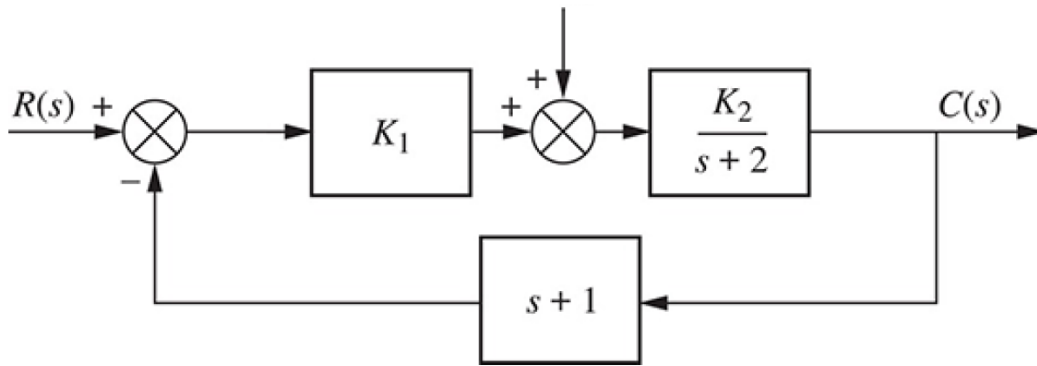
Why true?

from (26) $C = E G_1 G_2 + D G_2$, if $R=0$ then $E = 0 - C$, $C = -E$

$$\therefore -E = E G_1 G_2 + D G_2 \quad \therefore E = \frac{-D G_2}{1 + G_1 G_2}$$

如果遇到了非单位反馈，即 H 不是1，那么我们只需要在做input error analysis的时候找一个 G_e ，类似于下面提到的那样；对于分析disturbance error analysis，我们直接把 H 和controller的 G 相加，然后等效为上面图里的 G_1 就行。

$$D(s)$$



Let $C(s) = \bar{E} G_1 G_2 + D G_2$, $\bar{E}$ (error) = $R - C = -C$, junction

$$\bar{E}_j = 0 - CH = -CH, \quad \bar{E}_j = \bar{E} \cdot H$$

$$\therefore -\frac{\bar{E}_j}{H} = \bar{E}_j G_1 G_2 + D G_2$$

$$\therefore \bar{E}_j \left(G_1 G_2 + \frac{1}{H} \right) = -D G_2 \quad \therefore \bar{E}_j = \frac{-D G_2}{\left(G_1 G_2 + \frac{1}{H} \right)} = \frac{-D G_2 H}{(1 + G_1 G_2 H)}$$

$$\therefore \bar{E} = \frac{\bar{E}_j}{H} = \frac{-D G_2}{1 + G_2 G_1 H}$$

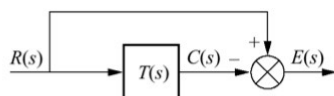
Steady-State Error and State Space

- We now consider how to evaluate steady-state error for a system represented in state-space.
- As we saw in Section 3.6 of the text, we can convert a single-input single-output state-space representation to an equivalent closed-loop transfer function using

$$T(s) = \frac{Y(s)}{U(s)} = \underline{C}(s\mathbf{I} - \underline{A})^{-1}\underline{B} \quad (34)$$

- In Diagram below, we have $E(s) = R(s) - C(s)$.
- We also have:

$$C(s) = R(s)T(s) \quad (35)$$



这一条是用state space的方法来算稳态误差。

Steady-State Error and State Space - II

- Combining the two we get

$$E(s) = R(s) - R(s)T(s) = R(s)[1 - T(s)] \quad (36)$$

- We thus have

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s E(s) \\ &= \lim_{s \rightarrow 0} s R(s)[1 - T(s)] \end{aligned} \quad (37)$$

- Substituting in for $T(s)$ gives

$$e_{ss} = \lim_{s \rightarrow 0} s R(s)[1 - \underline{C}(s\underline{I} - \underline{A})^{-1}\underline{B}] \quad (38)$$

做题步骤:

1. Steady state error的核心在于final value theorem, 即 $\lim_{s \rightarrow 0} sE(s)$, 然后用 $E(s) = \frac{R(s)}{1+G(s)}$ 。但是注意: 这个只适用于unity feedback system, 即 $H(s)=1$ 。
2. 判断输入类型, step input就是 $r(t) = Atu(t), R(s) = \frac{A}{s^2}$; ramp input是 $r(t) = Atu(t), R(s) = \frac{A}{s^2}$; parabolic input是 $r(t) = \frac{A}{2}t^2u(t), R(s) = \frac{A}{s^3}$ 。
3. 对于 unity feedback system, 我们有特殊公式。Step input的error是

$e_{ss} = \frac{A}{1+K_p}, K_p = \lim_{s \rightarrow 0} G(s)$ 。Ramp input是 $e_{ss} = \frac{A}{K_v}, K_v = \lim_{s \rightarrow 0} s * G(s)$ 。Parabolic input是 $\frac{A}{K_a}, K_a = \lim_{s \rightarrow 0} s^2 * G(s)$ 。

4. System type是 $G(s)$ 的分母的 s 项的次数，一般是0, 1, 2。

怎么处理nonunity feedback system:

Example 7.8 Steady-State Error for Nonunity-Feedback Systems

PROBLEM:

For the system shown in [Figure 7.16](#), find the system type, the appropriate error constant associated with the system type, and the steady-state error for a unit step input. Assume input and output units are the same.

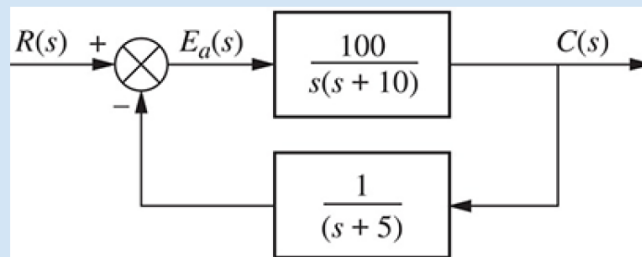


FIGURE 7.16 Nonunity-feedback control system for [Example 7.8](#)

1. 这是一个nonunity的系统，注意不能上来就判断系统的type，只有unity H system可以判断然后用快捷公式。
2. 我们假设存在一个 $G_e(s)$ ，让 $\frac{G_e(s)}{1+G_e(s)} = \frac{G(s)}{1+G(s)*H(s)}$ ，这样就能把这个 nonunity feedback system化简为unity feedback system，这样就可以确定系统的type然后计算。
3. 这个系统的 $G_e(s) = \frac{100*(s+5)}{s^3+15s^2-50s-400}$ ，这时候我们可以判断这是type 0 system，所以 $e_{ss} = \frac{1}{1+K_p}$ ， $K_p = \lim_{s \rightarrow 0} G_e(s) = 100 * \frac{5}{-400} = -\frac{5}{4}$ 。此时我们得到 $e_{ss} = \frac{1}{1-\frac{5}{4}} = -4$ 。